

Introduction to Data Analytics – Lifecycle, types, tools, dataset exploration.

What is Data Analytics?

Definition

Data Analytics is the process of collecting, organizing, cleaning, analyzing, and interpreting data to discover useful information, identify patterns, and support decision-making.

It helps organizations answer questions like:

What happened?

Why did it happen?

What will happen next?

What should we do?

Simple Example

An online shopping website records customer purchases.

Using Data Analytics, the company can find:

Best-selling products

Customer buying behavior

Monthly sales trend

Products with low demand

Future sales prediction

Why is Data Analytics Important?

Organizations use Data Analytics to

- Improve decision making

- Increase profit
- Reduce operational cost
- Improve customer satisfaction
- Predict future trends
- Detect fraud
- Optimize business performance
- Applications of Data Analytics
- Business
- Sales Analysis
- Customer Segmentation
- Profit Analysis
- Healthcare
- Disease Prediction
- Patient Monitoring
- Hospital Management
- Banking
- Fraud Detection
- Credit Score Analysis
- Risk Management
- Education
- Student Performance Analysis
- Attendance Monitoring
- Manufacturing
- Quality Control
- Predictive Maintenance
- E-commerce
- Product Recommendation
- Customer Behavior Analysis
- Government
- Population Analysis
- Traffic Management
- Public Health

Topic 2: Data Analytics Lifecycle

The Data Analytics Lifecycle is a structured process followed to analyze data effectively.

Step 1: Business Understanding

Understand the business problem.

Questions:

What is the objective?

What problem needs to be solved?

What are the expected outcomes?

Example:

A supermarket wants to increase sales.

Step 2: Data Collection

Collect data from various sources.

Sources include:

Databases

Excel Files

APIs

Websites

IoT Devices

Customer Surveys

Example:

Collect one year's sales data.

Step 3: Data Cleaning

Raw data usually contains:

Missing values

Duplicate records

Incorrect values

Inconsistent formatting

Cleaning improves data quality.

Example:

Remove duplicate customer entries.

Step 4: Data Exploration (EDA)

Analyze the data using statistics and visualizations.

Tasks:

Summary statistics

Correlation

Distribution

Trend analysis

Outlier detection

Common Charts:

Bar Chart

Pie Chart

Histogram

Scatter Plot

Line Chart

Step 5: Data Analysis

Apply analytical techniques.

Examples:

Sales trend analysis

Customer segmentation

Revenue analysis

Step 6: Data Visualization

Present insights using charts and dashboards.

Popular visualization tools:

Power BI

Tableau

Excel

Python (Matplotlib, Seaborn)

Step 7: Decision Making

Use insights to improve business performance.

Example:

Increase stock of high-demand products.

Data Analytics Lifecycle Diagram

Business Problem



Data Collection



Data Cleaning



Data Exploration (EDA)



Data Analysis



Visualization



Business Decision

Topic 3: Types of Data Analytics

There are four major types of Data Analytics.

1. Descriptive Analytics

Question Answered

What happened?

Purpose

Summarizes historical data.

Examples

Monthly Sales Report

Attendance Report

Profit Report

Example

Monthly Sales:

Month Sales

January ₹80,000

February ₹95,000

March ₹1,10,000

Observation:

Sales increased each month.

2. Diagnostic Analytics

Question Answered

Why did it happen?

Purpose

Finds the root cause of a problem.

Example

Sales dropped because:

Product shortage

Competitor discounts

Seasonal demand

3. Predictive Analytics

Question Answered

What is likely to happen?

Purpose

Uses Machine Learning and statistics to predict future events.

Examples:

Weather Forecast

Stock Price Prediction

Customer Churn Prediction

4. Prescriptive Analytics

Question Answered

What should we do?

Purpose

Provides recommendations.

Example:

Recommend increasing online advertising budget based on predicted sales trends.

Comparison Table

Type	Question	Example
Descriptive	What happened?	Sales Report
Diagnostic	Why did it happen?	Sales Drop Analysis
Predictive	What will happen?	Sales Forecast
Prescriptive	What should be done?	Marketing Recommendation

Topic 4: Data Analytics Tools

Different tools are used depending on the stage of analysis.

Microsoft Excel

Uses:

Data Entry

Sorting

Filtering

Pivot Tables

Charts

Advantages:

Easy to learn

Widely available

Suitable for beginners

SQL (Structured Query Language)

Purpose:

Retrieve and manage data stored in relational databases.

Example:

```
SELECT * FROM Sales;
```

Applications:

Data Retrieval

Data Filtering

Aggregation

Joins

Python

Python is the most popular programming language for Data Analytics.

Popular Libraries:

Pandas

Used for:

Reading datasets

Cleaning data

Data manipulation

Example:

```
import pandas as pd
```

```
df = pd.read_csv("sales.csv")
```

NumPy

Used for:

Numerical computation

Arrays

Mathematical operations

Matplotlib

Used for:

Data Visualization

Example:

```
import matplotlib.pyplot as plt
```

```
plt.plot([1,2,3],[5,6,7])
```

```
plt.show()
```

Seaborn

Used for:

Advanced statistical visualization

Power BI

Purpose:

Business Intelligence Dashboard

Features:

Interactive Reports

KPI Dashboard

Filters

Slicers

DAX

Data Modeling

Applications:

Sales Dashboard

HR Dashboard

Finance Dashboard

Tableau

Features:

Drag-and-Drop Visualization

Dashboards

Storytelling

Business Reporting

R Programming

Used mainly for:

Statistical Analysis

Machine Learning

Data Visualization

Google Looker Studio

Used for:

Cloud Dashboards

Google Analytics Reporting

Marketing Reports

Tool Comparison

Tool	Purpose
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Excel	Small datasets, reporting
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SQL	Database querying
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Python	Data cleaning, analysis, automation, ML
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Power BI	Business dashboards
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Tableau Data visualization

R Statistical analysis

Looker Studio Cloud reporting

Topic 5: Dataset Exploration

Dataset Exploration is the process of understanding the structure, quality, and characteristics of a dataset before analysis. This is also known as Exploratory Data Analysis (EDA).

What is a Dataset?

A dataset is a collection of related data organized in rows and columns.

Example:

Student_ID	Name	Age	Marks
101	Ali	20	85
102	Sara	21	90
103	John	19	78

Rows and Columns

Rows

Each row represents one record or observation.

Columns

Each column represents one feature or attribute.

Types of Data

Numerical Data

Examples:

Age

Salary

Marks

Categorical Data

Examples:

Gender

Department

City

Date and Time Data

Examples:

Order Date

Joining Date

Text Data

Examples:

Customer Review

Product Description

Basic Dataset Exploration Tasks

View the first few rows

Check dataset size

Identify column names

Check data types

Find missing values

Detect duplicate records

Generate summary statistics